Cobia habitat and distribution change with climate change predicted using a deep animal movement model MoE-SSM

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Abstract

Understanding species distributions requires integrating environmental suitability with movement, yet most species distribution models (SDMs) do not explicitly represent movement processes. Here, we develop a multiscale Mixture-of-Experts Step Selection Model (MoE-SSM) that couples CNN-derived habitat suitability with LSTM-derived movement dynamics to predict movement-

constrained cobia (Rachycentron canadum) occurrence across the Gulf of America. The framework integrates acoustic telemetry, mark–recapture, and satellite tracking to represent movement from local space use to larger-scale displacement and seasonal migration. Habitat suitability identified a broad environmental envelope, whereas movement constrained predictions toward locations consistent with observed behavior. Integrating both components produced the strongest overall discrimination (AUC = 0.961; average precision = 0.757). Under end-of-century climate change conditions (SSP5-8.5), habitat suitability shifted substantially from the western and central Gulf toward the eastern Gulf and Florida–Atlantic margin, while changes in movement-constrained occurrence were smaller and more spatially restricted. Our framework bridges SDMs and movement ecology, providing a transferable approach for predicting how highly mobile species may redistribute under environmental change.

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1. Introduction

Recent declines in cobia (*Rachycentron canadum*) landings in Northwest Florida have raised concerns about potential changes in migration and their consequences for recreational fisheries and management. Shifts in ocean temperature and other environmental conditions may alter cobia habitat use and seasonal movements, complicating stock assessment and management decisions based on historical distributions. Understanding how environmental change interacts with movement behavior is therefore important for anticipating future cobia distributions.

Species distribution models (SDMs) are widely used to predict species distributions by relating occurrence or abundance to environmental conditions such as temperature, depth, and productivity (Elith & Leathwick, 2009). However, environmental suitability does not necessarily translate into realized occurrence**,** because distributions also depend on accessibility, movement, biotic interactions, and other ecological processes (Soberón & Peterson, 2005; Soberón & Nakamura, 2009). This distinction is particularly important for highly mobile and migratory species, whose occurrence depends not only on environmental conditions but also on previous location, movement capacity, behavioral state, and seasonal dynamics. The movement ecology paradigm therefore views space use as the outcome of interactions among internal state, movement and navigation capacities, and external environmental conditions (Nathan et al., 2008). Habitat-based SDMs may consequently identify a broad environmental envelope without explicitly representing the movement processes determining which portions of that envelope are actually used.

Movement and habitat selection also operate across spatial and temporal scales that are difficult to represent with conventional ecological observations (Levin, 1992). This challenge is particularly relevant for migratory fishes, for which local movements, larger-scale displacement, and seasonal migration can respond differently to environmental variability. Step-selection functions provide a framework for incorporating these processes by comparing observed movement steps with available alternatives conditional on an individual's previous location (Fortin et al., 2005; Thurfjell et al., 2014). Integrated step-selection analysis further combines movement and habitat selection within a common inferential framework (Avgar et al., 2016). Thus, habitat and movement models address complementary questions: where are environmental conditions suitable, and where is an individual likely to occur given how it moves? This distinction is increasingly important under climate change because shifts in environmental

suitability need not translate directly into realized redistribution. Movement can constrain redistribution but can also enable mobile species to track changing climatic conditions, particularly in marine systems (Lenoir et al., 2020).

Representing movement across these scales requires observations from complementary technologies. Acoustic telemetry provides high-frequency information on local movements but is spatially constrained by receiver networks; mark–recapture data characterize displacement over longer intervals without resolving intervening trajectories; and satellite tracking captures broader and offshore movements but is often comparatively sparse. Integrating these heterogeneous observations therefore offers an opportunity to connect local movement, larger-scale displacement, and migration within a common framework. Deep learning provides additional flexibility for learning complex spatial and sequential relationships, with convolutional neural networks (CNNs) suited to spatial environmental patterns and recurrent architectures such as long short-term memory (LSTM) networks that can represent sequential movement dynamics.

Here, we develop a multiscale Mixture-of-Experts Step Selection Model (MoE-SSM) that integrates acoustic telemetry, mark–recapture, and satellite tracking observations to model cobia habitat selection and movement across spatial and temporal scales. The framework couples a CNN-based habitat component with an LSTM-based movement component, allowing environmental suitability and movement dynamics to be learned jointly across complementary observation scales. Step Selection Functions (SSFs) evaluate habitat selection by comparing observed movement steps with available alternatives conditional on an individual's previous location (Fortin et al., 2005; Thurfjell et al., 2014). Our MoE-SSM extends this concept within species distribution modeling by integrating heterogeneous movement observations and jointly representing habitat suitability and movement dynamics, thereby moving from predictions of environmental suitability toward a movement-constrained approximation of the realized niche. We use this framework to (1) evaluate whether incorporating movement improves prediction beyond habitat suitability alone, (2) determine how complementary observation sources contribute information across movement scales, (3) characterize seasonal cobia space use, and (4) project how habitat suitability and movement-constrained occurrence may redistribute under future climate conditions. By distinguishing changes in environmental suitability from their expression through movement, this framework provides a mechanistically informed approach for anticipating the redistribution of highly mobile marine species under environmental change.

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2. Method

2.1. Environmental predictors

We combined static and dynamic environmental predictors to enable the model to separate long-term habitat structure from short-term environmental variability (Figure 1). Static environmental predictors were derived from Bio-ORACLE (Biological Ocean Rasters for Analysis of Climate and Environment; www.bio-oracle.org), which provides global marine climatologies at approximately 5 arcmin (~9 km) spatial resolution. These layers represent long-term environmental averages and were treated as temporally invariant across all detections. A total of 25 raster channels were constructed from Bio-ORACLE variables, including sea surface temperature (θ_o), salinity (so), chlorophyll concentration (chl), dissolved oxygen (o_2), current

speed (sws), current direction (swd), phytoplankton carbon ($phyc$), bathymetry, and slope (Table 1). Multiple summary statistics (e.g., mean, minimum, range) were included across surface and depth profiles to capture vertical structure. The current direction was encoded using sine and cosine transformations.

Dynamic key oceanographic variables, including temperature, salinity, currents, sea level, and mixed layer depth, are averaged throughout the full water column (Table 2). These environmental predictors were derived from the global ocean reanalysis product GLORYS12V1 https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030, which provides eddy-resolving ocean fields at 1/12° spatial resolution (~8 km) and 50 vertical levels. Bio-ORACLE layers provide a climatological baseline describing typical environmental conditions, whereas dynamic predictors represent temporally varying ocean conditions at scales relevant to mesoscale processes in modeling cobia movement. This distinction allows the model to condition habitat selection on both persistent environmental gradients and transient oceanographic processes.

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2.2. Multi-scale MoE-SSM model

We implemented a mixture-of-experts (MoE) framework to integrate heterogeneous movement and habitat data sources across spatial and temporal scales (Figure 2). The model consists of two experts. In the MoE configuration, two experts were trained using acoustic telemetry (∼2 km resolution) and satellite telemetry (∼10 km resolution), respectively, with inputs including real 10-step movement sequences, seasonal encodings, and directional movement. The MoE structure enables experts to learn complementary ecological processes across different spatial scales. The acoustic expert captures fine-scale habitat selection within approximately 30 km, reflecting receiver spacing and localized environmental variation. In contrast, the satellite expert learns large-scale habitat associations over hundreds of kilometers, capturing broad oceanographic features that are associated with migratory behavior. This division of labor allows the model to integrate local habitat preferences with long-range movement drivers, providing a biologically interpretable multi-scale representation of space use.

The model employs full soft routing, such that although the gate is initialized to favor matching experts (e.g., satellite data weighted toward the satellite expert), all experts receive gradients from all data types during training. This allows partial knowledge transfer across experts while preserving specialization. Acoustic data provide large sample sizes and dense temporal coverage but are characterized by noisier directional estimates, allowing the LSTM to learn statistical relationships between movement features—such as speed, turning behavior, and seasonality—and directional tendencies. In contrast, satellite telemetry provides more accurate estimates of movement direction over longer spatial scales but is limited in sample size. By combining these complementary data sources, the model leverages the strengths of each: acoustic data support robust learning of feature–movement relationships, while satellite data constrain and anchor the absolute directional structure of migration. This integration enables the model to represent both local movement variability and large-scale directional patterns, improving its ability to capture multi-scale movement behavior.

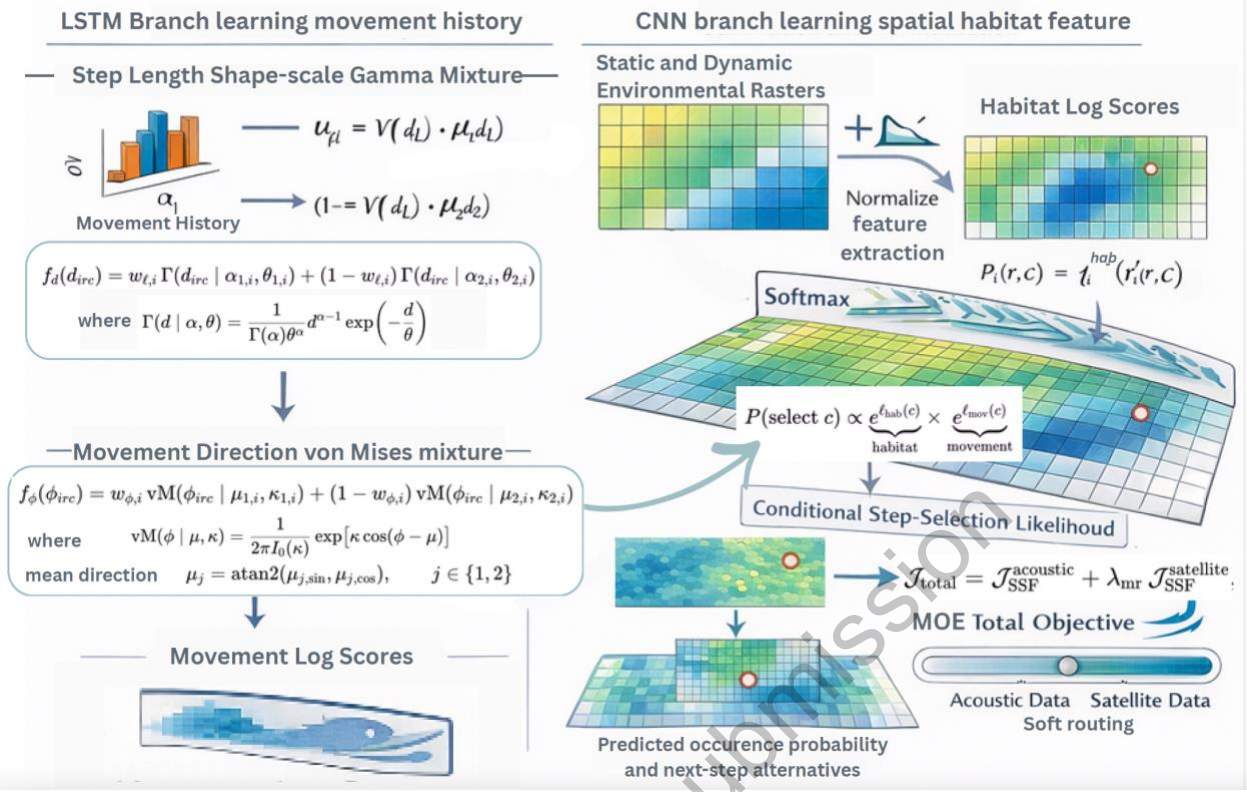


Figure 1: Diagram of soft routing MOE-SSM with integrated CNN-LSTM spatial and temporal branch to model cobia movements in the Gulf

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The gate is trained end-to-end using the final log likelihood of movement predictions via backpropagation, without an expert-level auxiliary objective (hard supervision from individual experts). Over-training, the gate is adapted by assigning higher weights to experts who improve predictive performance for each data source. In inference, predictions are generated using the acoustic gate configuration, allowing the model to selectively incorporate information learned from other data sources if beneficial. That is, the acoustic expert starts dominant (73%). During training on acoustic batches, the gate learns whether satellite experts add value for acoustic-scale predictions — if not, acoustic dominance grows, and the other shrinks further.

Each expert comprises a spatial (CNN) branch for habitat selection and a temporal (LSTM) branch for movement dynamics for the step-selection framework (SSF). The spatial branch of each expert is a six-layer 2-D convolutional neural network (CNN) that processes environmental raster windows centered on each candidate location. The temporal branch is an LSTM that encodes sequential movement features and outputs parameters defining a step length and a directional movement kernel. These parameters are evaluated across the spatial grid to produce a movement log-probability surface that reflects the reachability of each location given recent trajectory history. The model was therefore designed to answer the question: "Where is the right combination of good habitat and plausible movement?" with two experts. Models were trained using stochastic gradient-based optimization with a learning rate of 5×10-4 and weight decay of 1×10-5 to regularize model parameters. Dropout was applied to both the convolutional and recurrent components, with a dropout rate of 0.2 used in the CNN layers and as LSTM dropout to mitigate overfitting and improve generalization.

2.3. LSTM branch learning movement history

Kinematic variables and cyclic temporal movement features were derived for each step. The temporal branch LSTM encoded 10-step movement history sequences with seasonal context consisting of 14 kinematic variables and cyclic temporal features (including (step distance, normalized) velocity_ms (speed), turn_angle_rad (turning angle), sin_hour, cos_hour (time-of-day), sin_month, cos_month (seasonal pattern appended on-the-fly from the month column). Cyclic temporal encodings for hour and month were included using sine and cosine transformations (See Supplementary Materials Table S1 for a complete information of movement features). All features were normalized to the range $[0, 1]$ to account for differences in spatial and temporal resolution across data types.

By encoding absolute bearing in the input sequence and incorporating both kinematic features (e.g., speed, turning angle) and absolute directional information (sine and cosine of the bearing), the model can learn patterns that extend beyond single-step persistence. The LSTM predicts movement direction from the full trajectory history rather than as a local correction to the previous step. This enables the model to capture both complex movement behaviors and larger-scale trajectory structure, including directional reversals, gradual reorientation, and seasonal migration patterns. In addition, including temporal covariates enables the model to capture seasonal directional biases, such as shifts in migratory timing.

Table 3. Summary of movement features by data types

Feature	Acoustic Telemetry	MR (Mark–Recapture)	Satellite (Pop-up Tag)
MOVEMENT SCALE			
dist_m (median)	4,556 m (4.6 km)	12,700 m (12.7 km)	199,200 m (199.2 km)
dist_m (range p10–p90)	989 m – 72.5 km	~0 – 58 km	41 km – 759 km
dist_m (max, 99th pct cap)	591 km	92.9 km	1,797 km
Velocity (median)	12.2 cm/s	0.98 cm/s	~231 cm/s †
Bearing	Varies uniformly (std ±102°)	Computed per pair (e.g. 139.6° SE)	Computed per pair (e.g. −20.7° NNW)
Time interval (p10–p90)	1 h – 733 h	0 – 56 days	24 h (fixed)
Movement type	Short-range fine-scale daily	Coastal displacement over weeks	Long-range seasonal/migratory
MODEL INPUTS			
LSTM input	Real sequential steps (seq_len = 10)	Single step tiled × 10 ‡	Real sequential steps (seq_len = 10)
CNN pixel size	2 km/pixel	2 km/pixel	10 km/pixel
CNN grid radius	±100 km	±100 km	±500 km
MoE expert (source_type)	0 — acoustic	0 — acoustic (no dedicated expert) §	1 — satellite
DATASET			
N training samples	1,802 steps	181 pairs	406 pairs

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Feature	Acoustic Telemetry	MR (Mark–Recapture)	Satellite (Pop-up Tag)
N unique individuals	132	181 (each unique release)	~13 tags
Data source	Fixed receiver arrays	Physical tag & angler recapture	Wildlife Computers pop-up satellite
Labeling	Detection at real receiver	Mark → recapture location	Deployment → pop-up location

This formulation differs fundamentally from turning-angle (offset-based) models, which constrain predictions to local deviations from the most recent heading. However, this increased flexibility comes with greater data requirements. Learning absolute directional patterns reliably requires sufficient representation of trajectory variability across individuals and conditions. So, the acoustic expert's LSTM sees a genuine 10-step movement history from acoustic and satellite experts. The only difference from a satellite expert is the scale. Acoustic steps are meters to kilometers over hours, while satellite steps are tens to hundreds of kilometers over days.

The final hidden state from the last step is then passed through two linear layers with dropout and ReLU activation to define the movement kernel, with a bivariate distribution: a Gamma for step length and a von Mises for direction. Step length was modeled using a two-component Gamma mixture in shape–scale parameterization:

$$f_d(d) = w_\ell \cdot \Gamma(d \mid \alpha_1, \theta_1) + (1 - w_\ell) \cdot \Gamma(d \mid \alpha_2, \theta_2)$$

$$\Gamma(d \mid \alpha, \theta) = (1 / (\Gamma(\alpha)\, \theta^{\alpha})) \cdot d^{\,\alpha-1} \cdot \exp(-d / \theta)$$

where θ is the scale parameter (mean = $\alpha\theta$).

Movement directions were modeled using a two-component von Mises mixture (mean directions and direction density), mean directions predicted as full absolute bearings via sine–cosine outputs and recovered as:

$$\mu_j = \operatorname{atan2}(\mu_{j,\sin}, \mu_{j,\cos}), \quad j \in \{1,2\}$$

allowing unconstrained predictions in $[-\pi, \pi]$ and avoiding discontinuities at $\pm\pi$. yielding full absolute bearings in $[-\pi,\pi]$. This formulation allows the model to predict any direction without reference to the previous movement step.

The directional density is:

$$f_\varphi(\varphi) = w_\varphi \cdot vM(\varphi \mid \mu_1, \kappa_1) + (1 - w_\varphi) \cdot vM(\varphi \mid \mu_2, \kappa_2)$$

$$\text{where } vM(\varphi \mid \mu, \kappa) = (1 / (2\pi\, I_0(\kappa))) \cdot \exp[\kappa \cos(\varphi - \mu)]$$

Movement kernels were then converted into a movement log-probability surface $\ell\text{mov}(r,c)$ over a local 101 × 101 grid, describing: "given how this fish has been moving, how likely is it to end up at that distance and bearing?"

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2.3. CNN spatial branch learning habitat selection

In parallel, the spatial branch uses a 6-layer convolutional neural network (CNN) to process environmental raster layers together with scalar covariates broadcast across the grid, producing a habitat log-probability surface over the same window. The raster patch is centered at the current/mark position. Every pixel in the 101×101 window represents a potential next location, with its own Bio-ORACLE + CMEMS values (temperature, depth, chlorophyll, etc. at that spot). The target pixel is wherever the animal actually went next (recap position projected onto the grid). The CNN processes produce the habitat log-probability grid $\ell^{hab}(r,c)$for each of the 10,201 pixels. The loss pushes the score at the true destination pixel to be the highest. $\ell^{hab}(r,c)$ represents "how suitable is this environment for a cobia based on environmental rasters (SST, chlorophyll, bathymetry, etc.) at that candidate cell.

The CNN operates on 2-D spatial grids — every input must have a height and width dimension (H×W = 101×101). But some features, like time-of-day or sin/cos of time-of-day, are single numbers per detection — they have no spatial dimension. Therefore, they were broadcast as Scalar inputs (a 101×101 layer in which every cell contains the same value) and concatenated with the raster channels before entering the CNN. The CNN receives all channels stacked along the spatial channel dimension. For example, at midnight (sin_hour = 0), the kernel weights push the habitat map toward deeper offshore cells; at noon (sin_hour = 1), the same spatial rasters are used, but filtered differently because the time channel has changed. The scalar acts like a conditional switch — the same environmental rasters, but different habitat selection patterns depending on the time of day or season.

All convolutions use reflect padding (same kernel/stride/padding for all layers). Reflect padding prevents the CNN from learning spurious edge artifacts (e.g., always predicting low habitat suitability near window borders simply because zero-padded values resemble the deep ocean), which would bias occurrence predictions near the boundaries of the 202 km acoustic window or the 1,010 km satellite window. The raster window crops a continuous environmental field (e.g., SST, depth). The true values just outside the window are close to the values just inside — reflect padding approximates that continuity, while zero padding would introduce a fake discontinuity at the window edge.

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2.4. CNN-LSTM integration

The LSTM does not process spatial rasters, and the CNN does not process movement sequences. Instead, the LSTM models how the animal is likely to move based on recent trajectory history, while the CNN models where suitable habitat occurs in the surrounding landscape. These two components are combined additively in log-space and normalized via softmax to yield the *step-selection* SSF probability surface P(r,c), where the denominator sums over all available grid cells in the softmax window:

$$P(r,c) = \exp(\ell(r,c)) / \Sigma_{r',c'} \exp(\ell(r',c'))$$

such that every possible location around the animal is scored and is comparable to others. The scoring assigns a higher probability to the location the animal actually chose. The total log-score $\ell(r,c)$ reflects both habitat suitability and movement dynamics:

$$\ell(r,c) = \ell^{hab}(r,c) + \ell^{mov}(r,c)$$

The loss and training objective are:

$$J_{SSF} = -\Sigma_i \log P\,(r_i^{obs}, c_i^{obs})$$
$$J_{total} = J_{SSF}^{acoustic} + \lambda_{mr} \cdot J_{SSF}^{satellite}$$

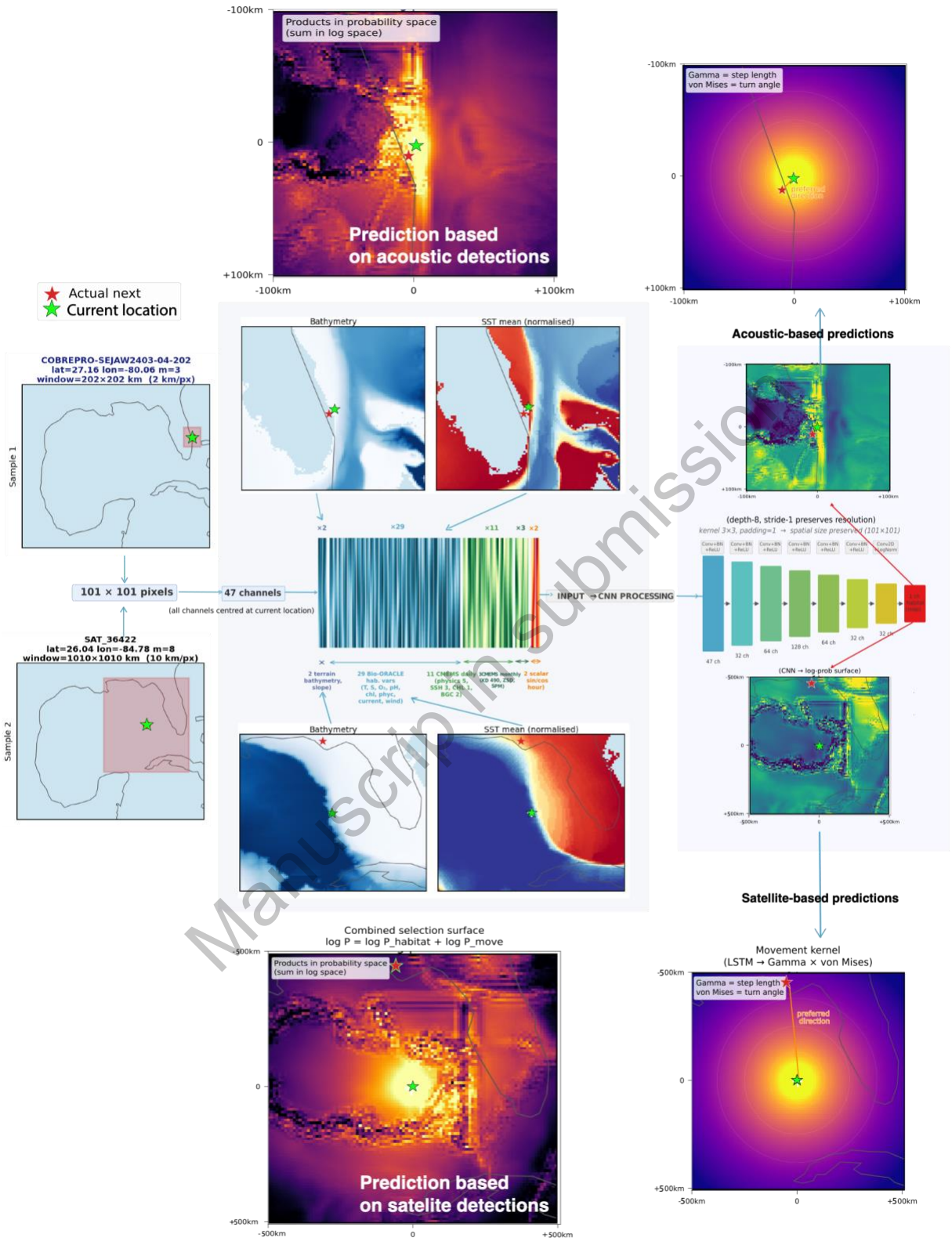
λ_{mr} downweights satellite data relative to acoustic data. Habitat and movement components are combined before normalization, and no separate loss or gating objective is used.

2.5. Multiscale Acoustic and Satellite training data

Training data for the MoE-SSF model were compiled from acoustic and satellite telemetry sources (Figure 2). Records with missing timestamps or spatial coordinates were removed, and temporal features were derived as month, day of year, and year. Coordinates were rounded to four decimal places to enable receiver-stay compression. For acoustic telemetry, only departure pings were retained so that consecutive observations represent true inter-receiver movements. For satellite telemetry, consecutive positions within 10 km were clustered to reduce GPS noise, with the final position in each cluster retained as the daily departure location. Movement steps were defined as transitions between consecutive locations, with acoustic steps exceeding 151 km removed to ensure compatibility with the 101 × 101 spatial grid (Figure 3). Acoustic data were split by individual into training, validation, and test sets (70/15/15; seed = 42), ensuring no individual appeared in multiple splits. Due to the limited sample size (n = 293 movement pairs), all satellite data were used exclusively for training. Satellite inputs were harmonized with the acoustic data structure with the same movement features, environmental predictors, and temporal scalers. All sequences were constructed as 10-step consecutive daily movements, providing consistent movement histories for LSTM-based modeling.

For each detection, the model produces a 101 × 101 spatial prediction window consisting of two log-probability surfaces corresponding to habitat selection and movement, each independently normalized via softmax so that probabilities sum to 1 within the window (Figure 3). The spatial resolution of these grids differs by data source: acoustic and mark–recapture (MR) data are represented at 2 km resolution (covering approximately 202 × 202 km), whereas satellite data are represented at 10 km resolution (covering approximately 1,010 × 1,010 km). This multi-resolution design preserves the scale of each data source and avoids the systematic exclusion of long-distance satellite movements; for example, the median step length of satellite pairs (~199 km) would exceed the spatial extent of a 2 km grid, leading to substantial data loss under a uniform resolution.

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Figure 2. Multiscale integration of neural networks and movement observations within the MoE-SSM framework for “next-step” predictions.

2.6. Occurrence probability prediction

We predicted the occurrence probability and pseudo-absence of cobia using a single-location (SDM-style) formulation and a paired, step-based (SSF-style) formulation, respectively, based on three data sources (acoustic telemetry, mark–recapture, and satellite telemetry). SDM formulation learns which environmental conditions distinguish observed locations from the broader environment, effectively characterizing species–environment associations consistently across the landscape. The SSF formulation learns which environmental conditions distinguish the selected next location from nearby, simultaneously available alternatives within the animal’s movement range, which represents conditional habitat selection rather than global habitat occupancy.

Single-location predictions were derived from individual detection records without movement history, including all acoustic detections, MR capture and recapture locations, and satellite positions (Figure 3). In this setting, temporal inputs to the movement model were set to zero, including movement sequences, scalar covariates, and bearing. As a result, the movement component produced a constant, uninformative kernel, and predictions were driven primarily by the habitat branch. Because the movement kernel is uninformative (constant), the model effectively estimates environmental availability:

$$P(select\ c) \propto exp(l_{hab}(c))$$

capturing global associations between species presence and environmental conditions. The habitat layer tells you where the environment is suitable, not where the animal could have gone. If we use environmental availability to generate pseudo-absences ignoring accessibility. A high-quality habitat patch 300 km away is not a valid pseudo-absence for a 50 km step, because it was never part of the available choice set. Thus, we used only single-location predictions to approximate habitat suitability or occurrence probability.

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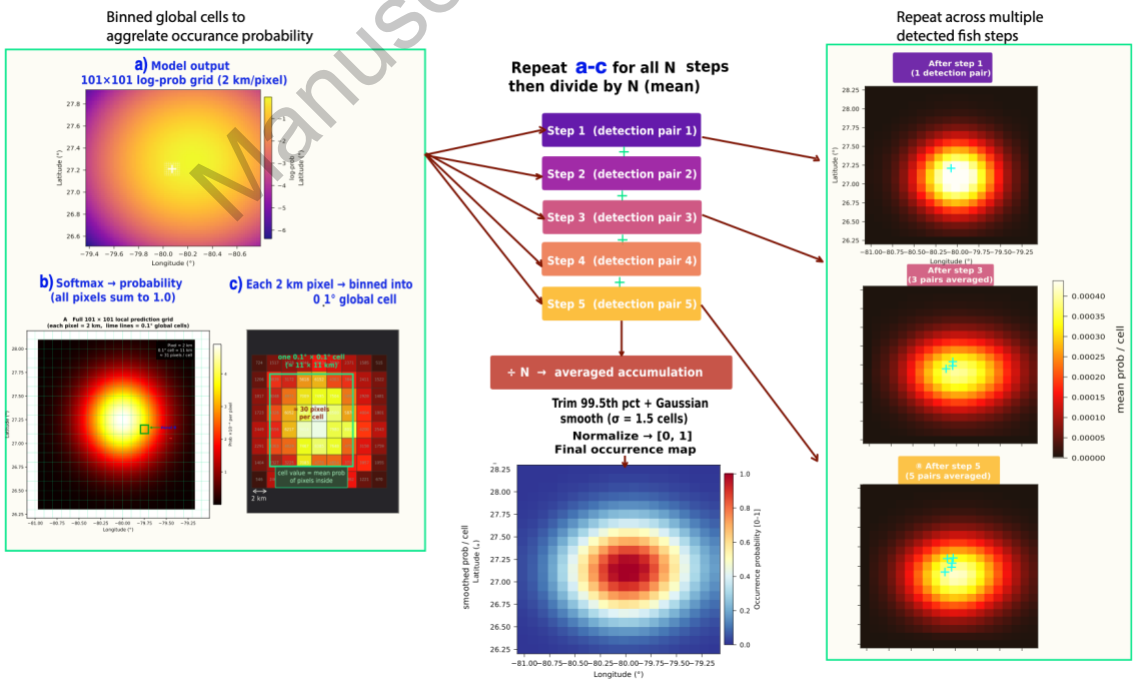


Figure 3. Conceptual framework for estimating movement-constrained occurrence probability in species distribution modeling.

To generate occurrence probability surfaces across the full study region, predictions were aggregated onto a common grid of 0.1° spatial cells. When aggregated across many detection locations, this fixed kernel is effectively "stamped" at each observation and summed over space:

$$M(c) = \sum_i \quad K(d(c, x_i)),$$

where $K(\cdot)$is the default movement kernel, and $x_i$are the detection locations. Because detections are spatially clustered (e.g., near receiver arrays), identical kernels accumulate and reinforce one another, producing spatial patterns that resemble a smoothed detection-density surface. Each cell received contributions from overlapping prediction windows, corresponding to approximately 25–30 acoustic pixels and typically one satellite pixel per cell. When multiple acoustic or MR predictions overlapped a cell, their values were averaged before aggregation. To ensure comparability across data sources with differing sample sizes, each source-specific prediction surface was normalized by its maximum value (scaled to a peak of 1.0) before combination. This normalization prevents sources with higher detection density from dominating the aggregated prediction and allows each data source to contribute equally to the final composite surface. It also ensures that color scales are comparable across sources, facilitating visual interpretation.

When movement inputs are absent in single-location predictions, the LSTM receives zero-valued sequences and therefore produces a fixed, non-zero output determined solely by its learned weights. The movement component, therefore, is uninformative, meaning it does not encode variation in movement processes across individuals or time. Consequently, when combined with the habitat component, the movement term acts as a spatial weighting function that is shaped by sampling effort rather than by movement dynamics. This can introduce bias in regions with uneven observation density, particularly for acoustic telemetry, where receiver coverage is spatially heterogeneous.

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2.7. Attribution of Climate-Driven Habitat Shifts

To quantify the contribution of individual oceanographic predictors to projected changes in cobia (*Rachycentron canadum*) distribution, we conducted a one-at-a-time (OAT) perturbation analysis (Morris, 1991; Hamby, 1994; Saltelli et al., 2004). Starting from present-day environmental conditions, each predictor group was independently replaced with its corresponding SSP5-8.5 end-of-century (2090–2100) projection (O'Neill et al., 2016; Riahi et al., 2017), while all remaining predictors were held at their present-day values. Predictor groups included potential temperature (θ_o), salinity (S_o), dissolved oxygen (O_2), pH, chlorophyll-a (Chl), and seawater current speed (sws; surface, minimum-depth, and maximum-depth layers). This design isolates the marginal model response associated with the projected change in each predictor while maintaining an otherwise present-day environmental background. Because predictors were perturbed individually, the analysis does not capture nonlinear interactions or joint responses among simultaneously changing environmental variables (Saltelli & Annoni, 2010; Pianosi et al., 2016).

For each OAT perturbation, the trained MoE-SSM was applied to the same set of observed acoustic-telemetry and satellite-tracking sequences used for the present-day reference prediction. Predictions were generated at the native resolution of each observation source (2 km pixel^{-1} for the acoustic expert and 10 km pixel^{-1} for the satellite expert) and subsequently aggregated to a common 0.1° grid. Perturbed and present-day predictions were processed using an identical normalization procedure. Within each source, non-zero grid values were trimmed at the 99.5th percentile and rescaled to [0, 1]. Source-normalized grids were then averaged, spatially smoothed using a Gaussian kernel ($\sigma = 1.5$ grid cells), and subjected to a final percentile-trim normalization to produce comparable composite surfaces bounded between 0 and 1.

For predictor k, the spatial change attributable to its SSP5-8.5 perturbation was calculated as

$$\Delta H_k(x) = \tilde{H}_{OAT,k}(x) - \tilde{H}_{present}(x)$$

where x denotes a 0.1° grid cell, and $\tilde{H}$ denotes the normalized model output. Positive values indicate that replacing baseline values of driver k with its projected future state increases predicted habitat use at that location; negative values indicate a decrease. We calculated each predictor's **scenario contribution** as the spatial mean of $|\Delta H_k|$ across all ocean grid cells, providing a scalar measure of the overall magnitude of model change associated with that predictor. For spatial comparison, we displayed predictor-specific change surfaces using a common diverging color scale, with symmetric limits defined by the 95th percentile of pooled $|\Delta H_k|$ values within each model component.

Because scenario contribution reflects both the magnitude of projected environmental change and model responsiveness, we additionally calculated standardized model sensitivity. For each predictor, forcing magnitude was calculated as the domain-mean absolute difference between future and present-day environmental values divided by the present-day spatial standard deviation of that predictor. Scenario contribution was then divided by this standardized forcing magnitude, yielding the model response per standard-deviation unit of projected environmental change (Hamby, 1994; Saltelli et al., 2004; Pianosi et al., 2016). Thus, scenario contribution identifies which projected environmental changes contribute most strongly to the SSP5-8.5 response, whereas standardized sensitivity identifies which predictors produce the strongest model response relative to the magnitude of their projected change.

OAT analyses were conducted separately for habitat suitability, movement probability, and integrated occurrence. Predictor-specific spatial changes were visualized using a common symmetric diverging scale within each model component, with limits derived from the pooled distribution of absolute changes across predictors, allowing direct comparison of the direction and magnitude of projected responses.

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3. Results

3.1. Current habitat, movement, and seasonal occurrence pattern predictions

The model decomposition revealed distinct contributions of habitat suitability and movement dynamics to predicted cobia occurrence (Figure 4a–c). Incorporating animal movement substantially altered the spatial distribution predicted from habitat suitability alone (Figure 4a–c). The habitat-only CNN predicted a broad and relatively continuous area of suitable conditions across the northern and eastern Gulf of America (Figure 4a). This surface represents the species' environmental envelope as learned from temperature, oceanographic, bathymetric, and other habitat covariates, but provides no information about whether an individual can plausibly reach a particular suitable location from its current position.

In contrast, the LSTM movement kernel produced a substantially more spatially constrained and heterogeneous probability surface (Figure 4b). By conditioning predictions on recent trajectory, speed, and direction, the movement component restricted candidate locations to areas that were plausible given the animal's preceding movement history. High movement probability occupied only a subset of the broader environmentally suitable region identified by the CNN, demonstrating that suitable habitat was substantially more extensive than movement-supported space use**.** Thus, environmental suitability alone was insufficient to reproduce the spatial concentration evident after accounting for movement.

Integrating habitat suitability and movement probability produced the most spatially resolved occurrence surface (Figure 4c). Areas received high occurrence probability only when they were supported by both favorable environmental conditions and the movement process, substantially narrowing the broad distribution suggested by habitat alone. This decomposition demonstrates that the two model components capture complementary ecological processes: habitat determines where an animal could potentially occur based on environmental conditions, whereas movement constrains where it can plausibly occur given its recent location and trajectory. The resulting integrated distribution therefore provides a closer representation of movement-informed realized space use than habitat suitability alone.

Seasonal predictions revealed pronounced variation in both the spatial extent and geographic distribution of occurrence (Figure 4d–g). Winter exhibited the smallest predicted extent **(**21,228 cells), with occurrence concentrated primarily in the north-central Gulf and a secondary concentration west of Florida (Figure 4d). The distribution expanded during spring **(**28,638 cells), when multiple high-occurrence areas emerged across the northern Gulf shelf and toward western Florida (Figure 4e). Spatial extent reached its annual maximum during summer **(**51,068 cells**)**, approximately 2.4 times the winter extent, accompanied by a broad high-occurrence region spanning much of the north-central and eastern Gulf and more widespread occurrence throughout the basin (Figure 4f). By fall, predicted extent declined to 43,849 cells but remained substantially greater than in winter and spring (Figure 4g). High-occurrence areas became more concentrated along the central-to-eastern Gulf, accompanied by a modest westward extension of predicted occurrence.

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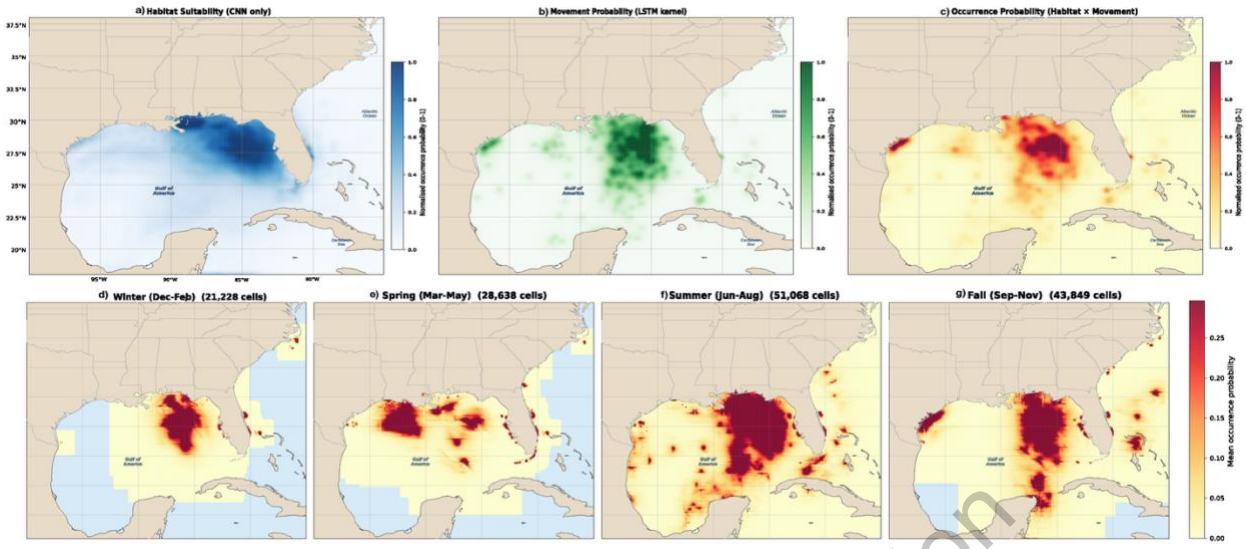


Figure 4. Predicted cobia occurrence probability in the Gulf of America from the Mixture-of-Experts hybrid step-selection function (MoE2 SSF) model. (a) Habitat suitability predicted by the convolutional neural network (CNN) branch; (b) Movement probability derived from the LSTM kernel; (c) Integrated occurrence probability (habitat suitability × movement probability) (d–g) Seasonal integrated occurrence predictions for Winter (Dec–Feb; 21,228 cells), Spring (Mar–May; 28,638 cells), Summer (Jun–Aug; 51,068 cells), and Fall (Sep–Nov; 43,849 cells). Cell counts indicate the number of grid cells with predicted occurrence above background.

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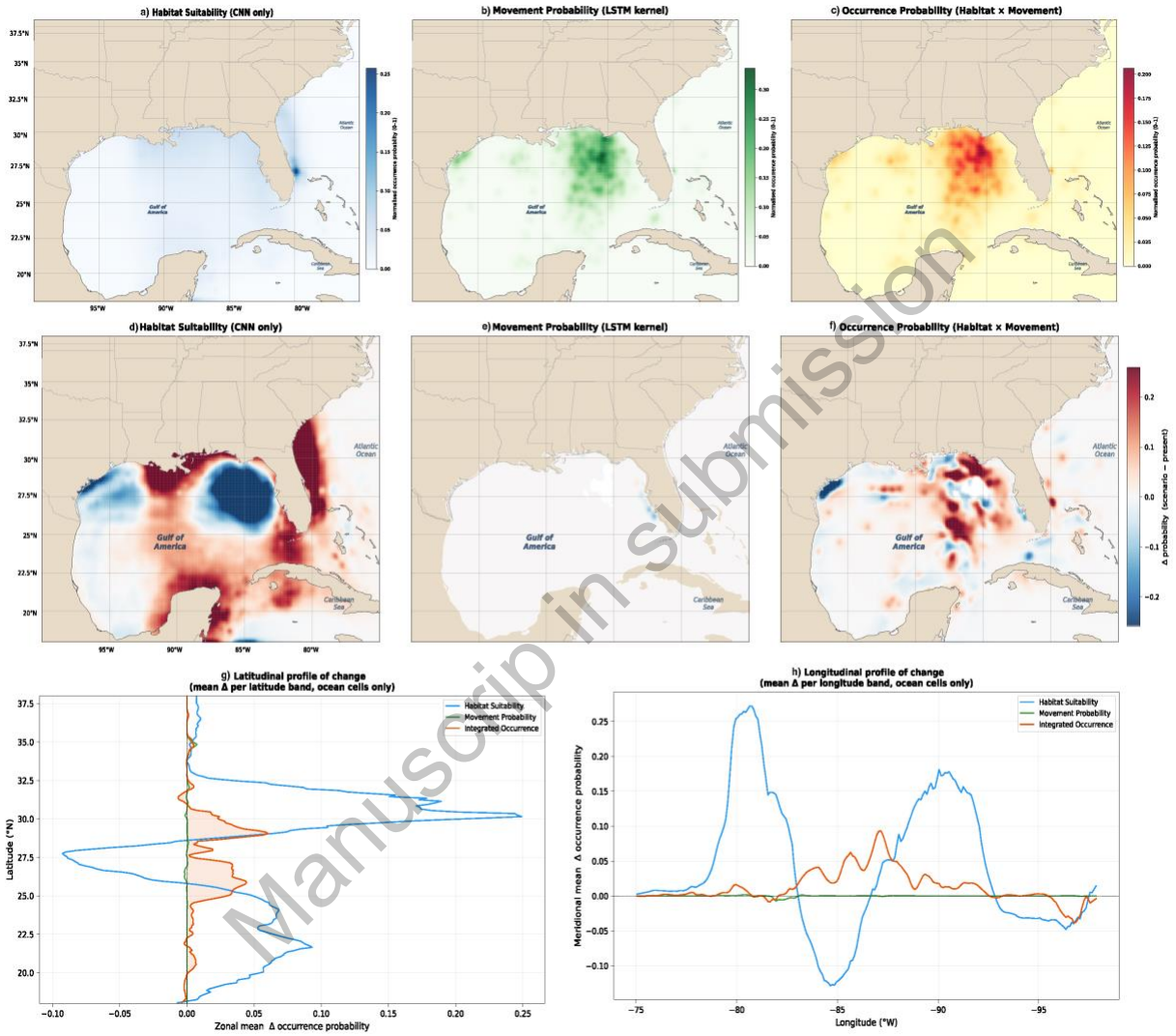
3.2. Projected changes under climate change

Under SSP5-8.5 (2090–2100), projected changes in cobia occurrence differed substantially between the habitat and movement components (Figure 5a–f). The CNN habitat-suitability branch, which incorporates environmental covariates, showed pronounced spatial redistribution. Future habitat suitability was concentrated primarily toward the eastern Gulf margin (Fig. 5a), while the change map showed widespread declines across the western and central Gulf (~88–95°W, 25–29°N) and increases along the eastern Gulf shelf and Atlantic margin near ~80°W (Figure 5d). In contrast, the LSTM movement branch showed negligible projected change across the domain (Figure 5b,e). This is because we applied movement relationships learned from contemporary tracking observations to the future scenario rather than projecting a future behavioral response independently.

The integrated occurrence projection reflected the contrasting responses of the two model components (Figure 5c,f). Projected declines were concentrated across the central and western Gulf, whereas more modest increases occurred along the eastern Gulf shelf and Florida–Atlantic margin. Because the movement component changed comparatively little, changes in integrated occurrence broadly followed the spatial redistribution of habitat suitability but were attenuated where movement probability was low. Thus, increases in environmental suitability did not translate uniformly into increases in predicted occurrence, because movement relationships constrained the contribution of newly suitable areas to the integrated distribution.

This mismatch between changing habitat suitability and more conservative movement-supported occurrence also illustrates an important ecological constraint on climate-driven

redistribution. Animals do not necessarily track newly suitable climatic conditions immediately or completely. For cobia, movement is likewise unlikely to respond to climate alone. Prey availability, bathymetric structure, and other local habitat features may further mediate whether newly climatically suitable areas become accessible and regularly used. These behavioral and ecological constraints may help explain why climate-driven changes in habitat suitability translated into smaller changes in integrated occurrence.



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3.3. Seasonal changes in habitat suitability and integrated occurrence under climate change

At the same time, the limited change in our movement component partly reflects the model structure: the LSTM learned movement relationships from contemporary tracking observations and applied these relationships to future projections. The integrated projections should therefore be interpreted as climate-driven changes in habitat suitability constrained by contemporary movement dynamics, rather than as a forecast in which movement behavior itself fully adapts to future climate conditions. Future changes in migration timing, dispersal, resource

tracking, or behavioral plasticity could either facilitate movement into newly suitable habitats or promote persistence in historically used areas, thereby altering the magnitude and timing of the redistribution projected here (Wrensford et al., 2025).

Directional profiles further demonstrated the geographic structure of projected redistribution (Figure 5g,h). Latitudinally, increases in habitat suitability were concentrated at approximately 29–32°N, whereas declines predominated at lower latitudes (~25–27°N; Figure 5g). Movement probability showed little change across the latitudinal gradient, while integrated occurrence followed the habitat response at a reduced magnitude. Longitudinally, the largest increases in habitat suitability occurred near 80°W, whereas the strongest declines occurred between approximately 88–95°W (Figure 5h). Integrated occurrence exhibited a similar but attenuated longitudinal pattern. Collectively, these projections indicate a geographic redistribution of modeled cobia occurrence, characterized by losses across the western and central Gulf and relative gains toward the eastern Gulf and Atlantic margin.

Given the pronounced seasonal movement and redistribution of cobia, we evaluated projected climate-driven changes separately for winter, spring, summer, and fall to determine whether future responses differed across the annual cycle (Figure 6a–j). Projected changes in habitat suitability under SSP5-8.5 were strongly season-dependent in both magnitude and spatial configuration (Fig. 6a–d). During winter, declines were concentrated primarily in the north-central Gulf (~28–30°N, 87–92°W), accompanied by more diffuse increases across portions of the western and southern Gulf (Fiure. 6a). Spring showed a similar but weaker concentration of habitat loss in the north-central Gulf (Figure 6b). Summer exhibited the most spatially concentrated decreases, centered farther east near the western Florida shelf (~27–30°N, 83–87°W), together with localized gains toward the Atlantic margin (Figure 6c). Fall was characterized by more diffuse decreases across the northern Gulf and broader areas of increasing suitability across the southern and western Gulf (Figure 6d). Thus, the location and magnitude of projected habitat change varied substantially among seasons rather than following a uniform geographic shift throughout the year.

Changes in integrated occurrence were consistently smaller in magnitude and more spatially localized than changes in habitat suitability (Figure 6e–h). Whereas habitat changes reached approximately ±0.4, integrated occurrence changes were largely within ±0.08 and were concentrated primarily along the northern Gulf shelf. Summer produced the most spatially heterogeneous integrated response, with interspersed areas of increasing and decreasing occurrence across the Louisiana–Mississippi–Florida Panhandle shelf (Figure 6g). During fall, increases were more apparent toward the northeastern Gulf, while decreases remained evident across portions of the north-central Gulf (Figure 6h). The reduced magnitude and spatial extent of integrated changes relative to habitat changes indicate that movement constrained the extent to which climate-driven changes in habitat suitability translated into changes in predicted occurrence**.**

Area-based comparisons further quantified this seasonal contrast ($|\Delta| > 0.02$; Figure 6i). The area experiencing increased habitat suitability exceeded the area experiencing decreases in all four seasons, with the largest gains occurring during summer (~45%) and fall (~41%). Habitat losses were greatest during summer (~15%) but were comparatively limited during fall (~3%). In contrast, changes in integrated occurrence occupied substantially smaller proportions of the modeled domain, with the largest areas of increasing occurrence occurring during summer and fall (~14% each). Thus, even in seasons characterized by extensive gains in environmental suitability, the corresponding expansion of integrated occurrence was considerably smaller.

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Probability-weighted centroids provided additional evidence of season-specific geographic redistribution (Figure 6j). Habitat-suitability centroids shifted eastward in all seasons, with the largest longitudinal displacement occurring during winter (~87°W to ~84.5°W). Integrated-occurrence centroids generally exhibited smaller eastward displacements, whereas movement-probability centroids changed minimally between present and future conditions. Latitudinal shifts were less consistent: spring showed a modest northward displacement, whereas summer and fall did not exhibit the same directional response. Projected redistribution was therefore consistently eastward in its longitudinal component but was not uniformly poleward across seasons.

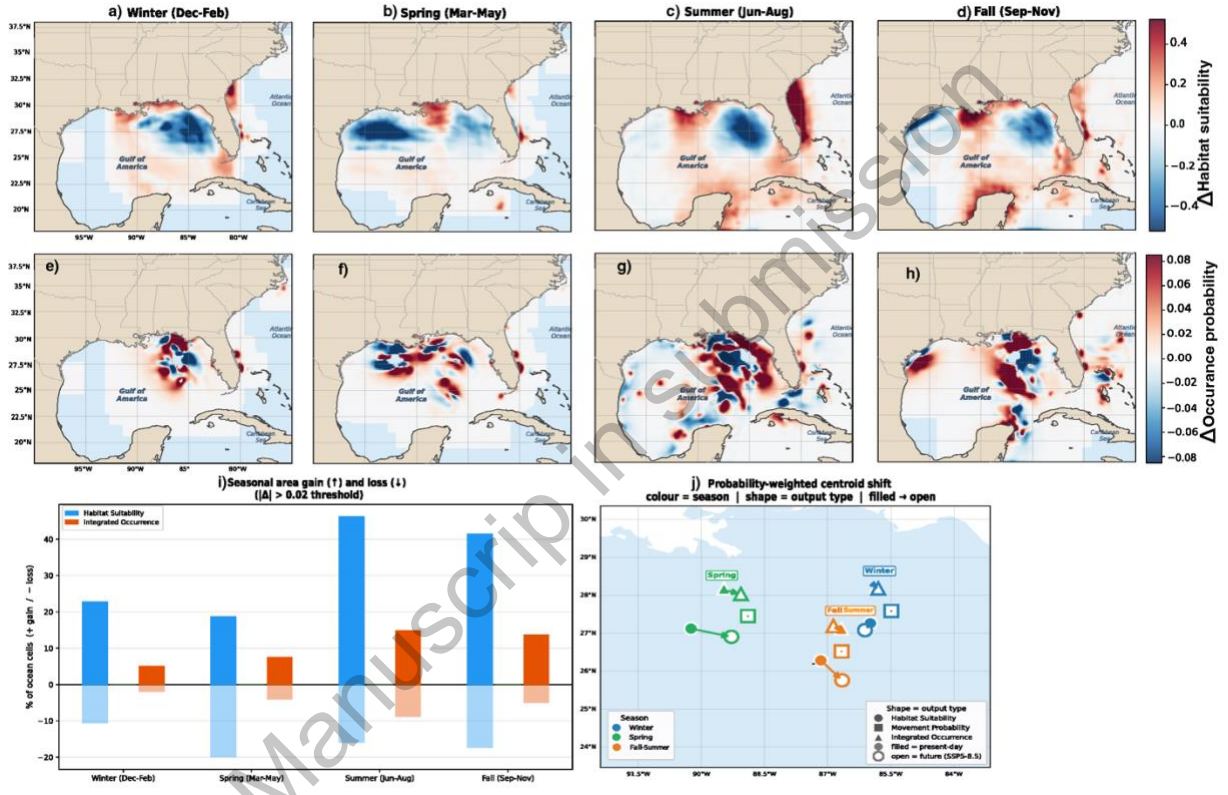


Figure 6. Projected seasonal changes in predicted cobia occurrence under the SSP5-8.5 climate scenario (2090–2100) relative to present-day conditions. (a–d) Projected change in habitat suitability (CNN branch; Δ = scenario − present) for Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug), and Fall (Sep–Nov). (e–h) Corresponding projected change in integrated occurrence probability (Habitat × Movement) for each season. Color scales are diverging: red indicates projected increase and blue indicates projected decrease. (i) Percentage of ocean grid cells with projected gain or loss exceeding an absolute threshold of 0.02 (j) Probability-weighted centroid positions for habitat suitability (circles), movement probability (squares), and integrated occurrence (triangles) under present-day (filled symbols) and future SSP5-8.5 (open symbols) conditions, colored by season. Arrows connect present to future centroids for each output type and season.

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3.4. Environmental drivers of projected change

One-at-a-time (OAT) sensitivity analysis revealed substantial differences between the environmental variables that contributed most to the SSP5-8.5 projection and those to which the model was most sensitive per unit forcing (Fig. X). For habitat suitability, pH produced by far the largest scenario contribution (0.0965), approximately five times that of dissolved oxygen (0.0211), the second-largest contributor. Contributions from potential temperature, salinity, chlorophyll-*a*, and seawater speed were smaller. Thus, pH was the dominant environmental contributor to projected habitat-suitability change under the SSP5-8.5 forcing represented in the model**.**

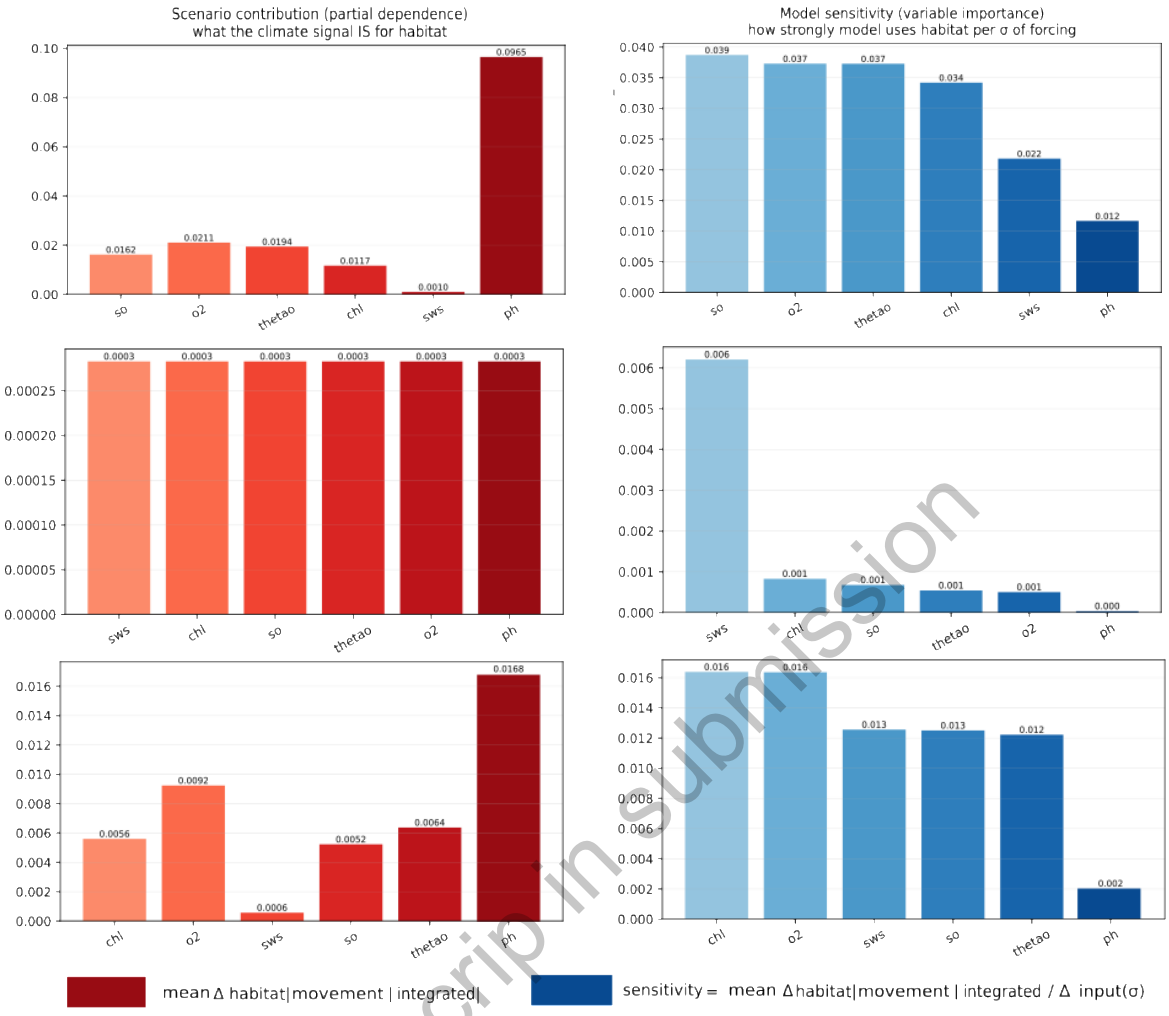
This dominance did not reflect particularly high model sensitivity to pH. When each variable was evaluated relative to an equivalent standardized change, habitat suitability was most sensitive to salinity (0.039), dissolved oxygen (0.037), and potential temperature (0.037), whereas sensitivity to pH was lowest (0.012). The contrasting rankings indicate that the large scenario contribution of pH arose primarily from the magnitude of its projected change under SSP5-8.5 rather than from unusually strong model responsiveness to pH. Conversely, salinity, dissolved oxygen, and temperature produced smaller scenario contributions despite stronger model responses per standardized unit of forcing.

Environmental contributions to movement probability were negligible compared with those for habitat suitability. Scenario contributions were approximately 0.0003 across all six environmental predictors, and standardized sensitivities were similarly small, although seawater speed produced the largest movement response (0.006). This result is consistent with the minimal future change in movement probability observed in the spatial and seasonal projections and indicates that, within the present model and perturbation experiment, environmental forcing contributed little to projected changes in the movement component.

The integrated occurrence response retained the environmental signal of the habitat component but at substantially reduced magnitude. pH remained the largest scenario contributor (0.0168), followed by dissolved oxygen (0.0092) and potential temperature (0.0064), whereas seawater speed contributed least (0.0006). In contrast, standardized model sensitivity was more evenly distributed among environmental predictors, with chlorophyll-*a* and dissolved oxygen producing the strongest responses (0.016 each), followed by seawater speed and salinity (0.013 each) and potential temperature (0.012); pH again produced the weakest standardized response (0.002). Thus, the dominant contribution of pH to projected integrated occurrence was associated primarily with the magnitude of scenario forcing rather than exceptional model sensitivity to pH.

Overall, the OAT analysis distinguished two complementary dimensions of environmental influence: scenario contribution identified which projected environmental changes contributed most to the SSP5-8.5 response, whereas standardized sensitivity identified the variables to which model predictions were most responsive per unit forcing**.** Their contrasting rankings demonstrate that the environmental variable contributing most strongly to a future projection is not necessarily the variable to which the model is most sensitive.

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Figure 7. *Cobia detections contributing at least one movement step and movement-kernel pseudo-absences by data source.*

3.5. Prediction performance

Figure 8a-c isolate the contributions of habitat and movement information to step-level prediction. The habitat-only CNN (A) produced spatially heterogeneous performance, with a median normalized rank of 2.3% of the 10,201-cell prediction grid and an AUC-ROC of 0.84. Adding the LSTM movement kernel (B), which incorporates recent trajectory, speed, and turning behavior, reduced the median normalized rank to 0.2% and increased AUC-ROC to 0.94, with 72.1% of observed steps ranked within the top 1% of grid cells. The integrated model (C) provided a further, more modest improvement (AUC-ROC = 0.96), indicating that habitat information adds discrimination after the movement kernel has constrained the set of geographically plausible destinations. Together, these results show that recent movement history provides the dominant signal for next-step prediction, while habitat acts as a complementary spatial constraint.

Spatially, the habitat-only model performed most poorly along the Texas–Louisiana shelf and in offshore areas, while performance was generally stronger in the eastern Gulf and along the Florida Atlantic coast. Incorporating the movement kernel substantially reduced these regional disparities. However, a persistent cluster of poorly ranked steps remained on the northwestern Gulf shelf across all three panels, suggesting that movements in this region are less well represented by the current environmental covariates and learned movement dynamics.

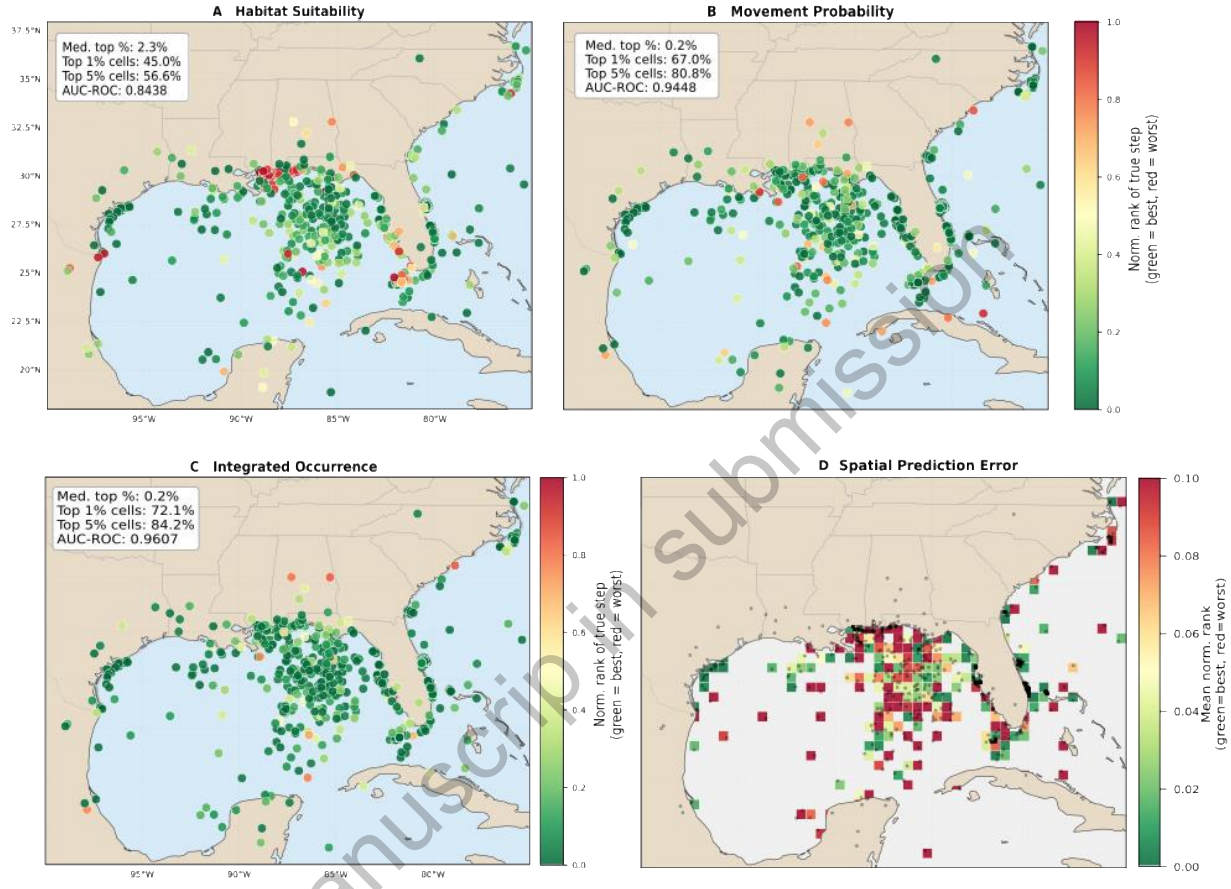


Figure 8. Spatial distribution of model prediction performance

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Aggregating integrated-model performance into 0.5° grid cells confirms the geographic structure evident in Panels A–C (Figure 8). Prediction performance was strongest in the northeastern Gulf and along the Florida Atlantic coast. In contrast, the most coherent region of elevated error occurred along the Louisiana–Texas shelf, reinforcing the interpretation that this area represents a spatially systematic model limitation rather than a collection of isolated prediction failures. Poorly predicted cells elsewhere in the southern and central Gulf were more spatially isolated and should be interpreted cautiously, particularly where individual grid cells contained relatively few observations. Overall, prediction errors were geographically structured but localized, and their spatial distribution was consistent with the quantitative improvements observed from habitat-only to movement-informed and integrated predictions (More model evaluation procedures are in Supplementary Materials).

Model evaluation metric curves demonstrated a substantial improvement in predictive performance when movement information was incorporated into habitat-based predictions (Figure 9a–e). All three model outputs discriminated observed next-step locations from randomly sampled alternatives substantially better than chance, but performance differed markedly among components (Figure 9a). Integrated occurrence achieved the highest discrimination (AUC = 0.961), followed closely by movement probability (AUC = 0.945), whereas habitat suitability alone performed substantially lower (AUC = 0.844). The steep rise of the integrated and movement ROC curves at low false-positive rates further indicated that observed locations were preferentially assigned high relative probabilities once movement information was incorporated.

Precision–recall analysis produced an even stronger separation among model components under the highly imbalanced evaluation design of one observed step to 99 random locations (Figure 9b). Integrated occurrence achieved the highest average precision (AP = 0.757), followed by movement probability (AP = 0.692), while habitat suitability alone achieved AP = 0.378. Thus, the improvement associated with movement was maintained when evaluation emphasized the model's ability to identify observed locations among a large number of available alternatives.

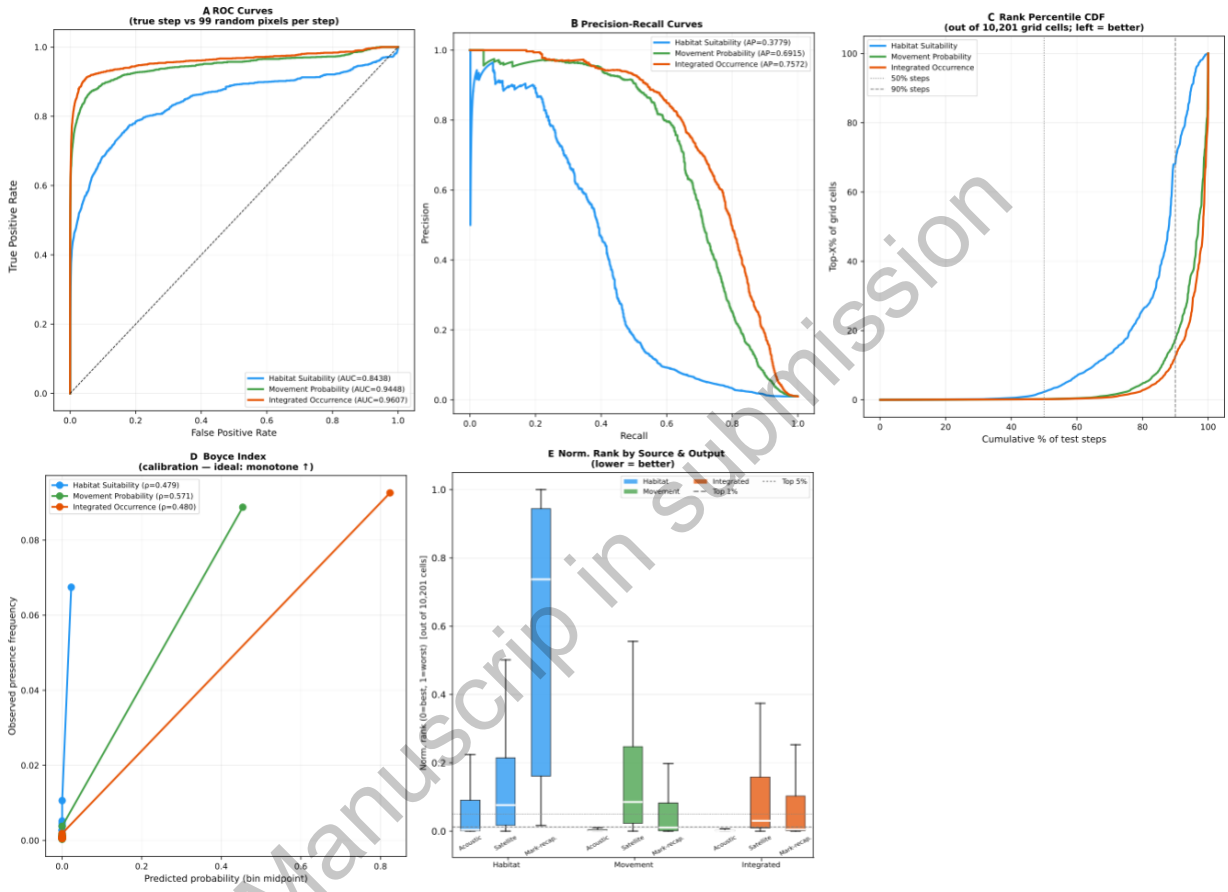
Ranking observed locations against the complete 10,201-cell prediction grid provided a more stringent spatial evaluation (Figure 9c). Integrated occurrence and movement probability consistently placed observed next-step locations within a small fraction of the available grid, with their cumulative rank distributions strongly concentrated toward zero. Habitat suitability produced substantially broader ranks, indicating that environmental suitability alone identified a much larger set of plausible locations. Together, these results show that habitat suitability defined the broader environmental envelope, whereas movement information substantially narrowed that envelope toward locations consistent with observed animal trajectories. Integrating the two components provided the strongest overall discrimination, indicating that habitat information added predictive power after movement constraints were imposed.

The Boyce index provided a complementary assessment of consistency between predicted values and observed use (Fig. 9d). All three components showed positive associations between predicted values and observed frequency, but movement probability achieved the highest Boyce index ($\rho = 0.571$), compared with integrated occurrence ($\rho = 0.480$) and habitat suitability ($\rho = 0.479$). Thus, although integrated occurrence provided the strongest discrimination and ranking performance, movement probability showed the strongest monotonic correspondence between predicted probability and observed use. This pattern is consistent with a trade-off between discrimination and calibration. The movement kernel's spatially concentrated probability surface produces a sharp gradient between predicted bins and observed use, whereas integration with the more diffuse habitat-suitability component broadens the probability surface in ways that improve step-level discrimination without strengthening the bin-level monotonic relationship.

Performance varied substantially by data source. For habitat suitability, acoustic steps achieved the most favorable ranks (median norm. rank ≈ 0.08), while satellite steps showed markedly poor habitat ranks (median ≈ 0.70). Mark-recapture performance was intermediate. Movement probability achieved near-optimal ranks for acoustic steps (median close to the top 1% threshold) and improved substantially over habitat suitability for satellite and mark-recapture steps. Integration further reduced ranks for acoustic steps to near the top 1% line and partially recovered satellite performance relative to habitat alone. Mark-recapture integrated ranks were intermediate. The persistent gap between acoustic and satellite ranks across all three output types indicates that satellite steps were systematically more difficult to predict than acoustic steps

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regardless of which model component was evaluated. Overall, the evaluation demonstrates that movement information accounted for the largest improvement over habitat-only prediction, while habitat and movement together provided the strongest overall localization of observed cobia occurrence. These complementary results support the central premise of the framework: environmental suitability identifies where cobia could occur, but movement dynamics are critical for resolving where an individual is likely to occur at a particular step; integrating both processes yields the most informative prediction of realized space use.



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Figure 9. Evaluation of the MoE-SSF model across three output components — habitat suitability (CNN; blue), movement probability (LSTM kernel; green), and integrated occurrence (Habitat × Movement; orange) — for 1,568 test steps across acoustic telemetry, mark–recapture, and satellite tracking data sources. (A) Receiver operating characteristic (ROC) curves (B) Precision–recall curves for the same contrast (C) Cumulative distribution of normalized rank percentile across all 10,201 prediction grid cells (D) Boyce index: observed presence frequency as a function of predicted probability bin midpoint for each output component (E) Normalized rank by data source (acoustic, satellite, mark–recapture) and output type

4. Discussion

4.1. Integration of neural networks

The broad habitat-suitability surface identified by the CNN is consistent with previous evidence that cobia distributions are structured by oceanographic conditions and continental-shelf environments and vary seasonally (Marín-Enríquez et al., 2022). Our results extend this habitat-based perspective by showing that incorporating movement substantially narrows and reorganizes environmentally suitable space. Across evaluation metrics, movement improved localization of observed positions, while habitat–movement integration produced the strongest overall discrimination. Thus, the CNN and LSTM captured complementary dimensions of space use: the CNN characterized environmental suitability, whereas the LSTM conditioned occurrence on empirically learned movement behavior.

This decomposition bridges habitat-based distribution modeling with movement approaches that explicitly condition spatial selection on the movement process (Nathan et al., 2008; Avgar et al., 2016). We therefore interpret the integrated output as **movement-constrained occurrence**, which more closely approximates realized space use than habitat suitability alone without equating it with the realized niche (Hutchinson, 1957; Soberón & Peterson, 2005). Although observed movement may implicitly reflect some ecological interactions, unmodeled processes such as predator–prey dynamics, conspecific attraction, and social behavior may further shape realized distributions (Soberón, 2007; Wisz et al., 2013).

4.2. Integration of complementary data sources through MoE

The three observation sources resolve cobia movement across a continuum from local movement to larger-scale displacement and seasonal migration. Animal movement is inherently hierarchical (Kays et al., 2023), and different tracking approaches provide complementary information across these scales (Harcourt et al., 2019). Acoustic telemetry resolves high-frequency local movements within receiver networks; mark–recapture captures displacement over longer temporal and spatial intervals but not intervening trajectories; and satellite tracking provides information on large-scale, offshore, and migratory movements. This multi-scale integration is particularly relevant for cobia, which exhibit both localized space use and long-distance seasonal movements (Streich et al., 2026). The MoE framework therefore integrates complementary information across movement scales that no single observation source adequately represents.

Source-specific performance also revealed limitations. Acoustic observations were localized most accurately, satellite observations were more difficult to predict, and mark–recapture was generally intermediate. Differences in sampling frequency, spatial coverage, positional resolution, and observation process can influence the movement scales resolved by tracking systems and subsequent ecological inference (Katzner & Arlettaz, 2020; Cullen et al., 2024). In our model, mark–recapture provided less sequential movement information, while the relatively small satellite sample limited learning of large-scale movements. Thus, multi-source integration broadens inference from local space use to migration, but source-specific observation structure and sampling imbalance remain important limitations.

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4.3. Movement-mediated climate redistribution

Climate projections demonstrate the importance of habitat–movement integration under environmental change. Under SSP5-8.5, habitat suitability underwent pronounced geographic and seasonal redistribution, whereas changes in integrated occurrence were substantially smaller. Thus, environmental change altered the geographic template of suitable habitat more strongly than movement-constrained occurrence within the present framework.

Movement can both constrain and facilitate redistribution. Site fidelity, spatial familiarity, life-history stage, resource distributions, and habitat structure can influence whether animals remain in historically occupied areas or use newly suitable environments, whereas behavioral flexibility can facilitate tracking of changing conditions (Abrahms et al., 2021; Merkle et al., 2022; Wrensford et al., 2025). Marine species can track shifting climatic conditions relatively effectively (Lenoir et al., 2020), making movement a dynamic mediator rather than simply a constraint. For cobia, documented seasonal fidelity and repeated use of continental-shelf habitats (Streich et al., 2026) suggest plausible mechanisms for such mediation, although site fidelity, prey availability, and bathymetric constraints were not independently tested here.

Importantly, the LSTM was learned from contemporary tracking observations. Its limited projected change should therefore not be interpreted as evidence that future cobia movement will remain unchanged. Climate change may alter movement routes, timing, speed, and resource tracking (Jonsen et al., 2019; Abrahms et al., 2021). Our projections are consequently best interpreted as future changes in habitat suitability filtered through contemporary movement dynamics**,** rather than fully dynamic forecasts of future movement. Future models that allow movement behavior to respond explicitly to projected environmental conditions could represent both changing habitat and behavioral adaptation simultaneously.

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4.4. Environmental drivers of projected redistribution

OAT analysis distinguished the environmental variables contributing most to the SSP5-8.5 projection from those to which the model was most responsive per standardized unit of forcing. pH produced the largest scenario contribution to habitat change but the lowest standardized sensitivity, whereas salinity, dissolved oxygen, and temperature produced stronger responses per unit forcing. Future projections therefore depend jointly on the magnitude of environmental change and model responsiveness to that change (Saltelli et al., 2004; Pianosi et al., 2016). The dominant pH contribution should consequently not be interpreted as evidence that cobia are inherently more sensitive to acidification; rather, its large projected change under SSP5-8.5 generated a strong scenario contribution despite relatively weak standardized model sensitivity.

The ecological mechanism underlying this pH association remains uncertain. Correlative models cannot distinguish direct physiological responses from associations with covarying environmental conditions (Elith & Leathwick, 2009; Dormann et al., 2012), and projected late-century pH extended beyond the contemporary training range, requiring extrapolation into novel environmental conditions (Elith et al., 2010). Mechanistic modeling and experimental evidence

will therefore be needed to determine whether pH directly constrains cobia habitat or partly represents broader oceanographic change.

4. Conclusion

This study demonstrates that integrating movement dynamics with habitat suitability provides a more realistic representation of species space use than environmental suitability alone. Habitat models identified the broader environmental envelope associated with cobia occurrence, whereas movement information constrained this envelope to locations consistent with observed movement behavior. Across complementary evaluation metrics, incorporating movement substantially improved the localization of observed cobia positions, while integrating habitat and movement provided the strongest overall discrimination. By combining acoustic telemetry, satellite tracking, and mark–recapture observations within a Mixture-of-Experts framework, the approach integrated complementary information across spatial and movement scales, providing a unified representation of cobia space use across the Gulf of America.

This distinction between environmental suitability and movement-constrained occurrence was also important under climate change. Under SSP5-8.5, projected habitat suitability showed substantial geographic and seasonal redistribution, whereas corresponding changes in integrated occurrence were smaller and more spatially constrained. Seasonal analyses further showed that redistribution varied across the annual cycle rather than following a uniform directional shift. OAT sensitivity analysis demonstrated that the environmental variables contributing most strongly to future projections were not necessarily those to which the model was most sensitive per unit forcing, highlighting the importance of distinguishing the magnitude of environmental change from model responsiveness to that change. Because the movement component was learned from contemporary tracking observations and changed little under future forcing, these projections are best interpreted as future environmental change filtered through contemporary movement dynamics, rather than as fully dynamic forecasts of future cobia movement and distribution.

Future development should focus on addressing sampling and observation biases among tracking sources, expanding offshore representation and balancing data sources through targeted satellite tracking, and developing standardized approaches for aggregating individual movement processes into population-level spatial predictions. Incorporating environmentally conditioned movement dynamics would further allow future movement behavior to respond explicitly to projected environmental change, while principled movement-informed availability or pseudo-absence sampling could improve grid-based inference. Together, these advances would strengthen the transferability of movement–habitat models across species, observation systems, and changing environmental conditions. More broadly, movement-informed habitat modeling provides a path from maps of where environments may become suitable to predictions of where mobile species can realistically redistribute, offering a more mechanistically informed and management-relevant basis for anticipating species responses to environmental change.

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